Blockchain-Enhanced Random Access in Untrusted Wireless Networks: An Evolutionary Game Analysis

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***Abstract*—The sixth-generation (6G) wireless network will bring major advances in ubiquitous connectivity, support massive device deployments, and open up new opportunities for diverse and decentralized Internet of things (IoT) applications. However, it also raises inherent trust issues, as IoT devices controlled by various independent entities cannot fully trust each other, thereby complicating secure and efficient random access control. To address this challenge, existing research turns to blockchain technology as a promising secure and trusted solution. This paper aims to explore and analyze blockchain-enhanced random access management from a game-theoretic perspective. We first investigate the well-known rogue's dilemma during random access process, where selfish devices exploit access protocols and reduce system fairness. Next, we construct a two-device mixed-strategy game to illustrate the strategic interaction between honest and selfish IoT devices. To capture realistic behaviors under bounded rationality, we extend the analysis to a multi-device scenario with evolutionary game theory and replicator dynamics. Numerical results show that unregulated selfish behavior severely reduces IoT network efficiency. In contrast, blockchain-based penalties effectively discourage malicious actions and promote stable and efficient access equilibria.**

***Index Terms*—Blockchain, evolutionary game, random access, wireless communication.**

I. Introduction

As a crucial application scenario for next-generation communication systems, the Internet of things (IoT) has experienced rapid growth in connected devices and significantly impacted critical areas such as vehicular networks, low-altitude networks, smart cities, and healthcare [1], [2]. With the extensive deploym ent of fifth-generation (5G) networks and the ongoing development of sixth-generation (6G) technologies, IoT connectivity is becoming more diverse, ubiquitous, and complex [3], [4]. In practical settings, IoT networks usually involve numerous devices produced by different manufacturers, managed by distinct service providers, and belonging to separate trust domains. Consequently, ensuring secure collaboration, efficient resource sharing, and trusted access among these heterogeneous entities has emerged as a significant challenge and key bottleneck for the continued advancement of IoT systems [5].

To address these challenges, researchers have proposed integrating blockchain technology into IoT access procedures [6], [7]. Since its introduction through Bitcoin in 2008, blockchain technology has attracted widespread attention [8]. As a distributed ledger, blockchain provides a reliable approach to building trust in digital environments by leveraging its core characteristics of transparency, immutability, and collective maintenance [9], [10]. Blockchain thus offers a secure and transparent solution for random access control in untrusted IoT environments. Particularly, hash access, a blockchain-enhanced random access protocol, has emerged as a promising approach for trusted random access in IoT systems [6]. By integrating blockchain consensus with hash competition mechanisms, hash access provides a trustworthy paradigm for managing access conflicts and resource contention in multi-device, untrusted environments [11]. Its robust security guarantees and resilience to malicious actions make it particularly suitable for the high-density, heterogeneous device connectivity expected in future 6G scenarios.

Despite these promising developments, the benefits of integrating blockchain into IoT random access protocols are primarily discussed qualitatively rather than rigorously demonstrated through mathematical analysis. Several critical issues remain insufficiently addressed in the existing studies. First, the behaviors of selfish nodes, including the associated costs and benefits of violating established access rules, are inadequately

characterized. Second, the long-term impacts of such selfish behaviors on the fairness and efficiency of overall system are not fully understood. Third, appropriate system configurations capable of effectively mitigating selfish behaviors in untrusted IoT environments remain unexplored.

In this work, we provide a game-theoretic analysis of selfish device behaviors and the performance of blockchain-enhanced random access in untrusted IoT environments. The main contributions of this paper are summarized as follows.

- We characterize the behavior of selfish devices and analyze the rogue's dilemma in blockchain-enhanced random access, where selfish nodes violate backoff rules for unfair gain. We also quantify the incentives and outcomes of those selfish actions through game theory.
- We construct a two-device mixed-strategy game model to capture interactions between honest and selfish nodes, and then extend it to a multi-device scenario using evolutionary game and replicator dynamics to derive evolutionarily stable strategies (ESS).
- Numerical results validate our findings and demonstrate that blockchain-based penalties can effectively deter selfish behavior, promote stable access strategies, and improve overall network efficiency in untrusted IoT environments.

The remainder of this paper is organized as follows. Section II provides a brief overview of the blockchain-enhanced random access paradigm. Section III analyzes the rogue's dilemma caused by selfish behavior of rogue devices. Section IV establishes a mixed-strategy game model and an evolutionary game model to derive the stable and efficient strategies for both honest and rogue nodes in untrusted IoT environments. Section V shows simulation results and discussions. Finally, Section VI concludes this paper.

II. Blockchain-Enhanced Random Access

Traditional random access management allows multiple IoT devices to compete for a shared channel to transmit data. A transmission is successful only when a single device transmits within a given time slot, while simultaneous transmissions result in collisions and lead to packet loss [12], [13]. In the classic additive links on-line Hawaii area (ALOHA) protocol, IoT devices experiencing collisions perform random backoff and attempt retransmission after a randomly selected delay [14]. However, as the density and operators of IoT devices increase, these ALOHA-based protocols face several limitations. First, the absence of secure and trusted coordination mechanisms impedes efficient access management across heterogeneous subnets. Second, selfish behaviors such as repeated retransmissions without appropriate backoff lead to network congestion. Third, the lack of authentication mechanisms undermines the reliability of data delivery to the core network, especially in the presence of interference or forwarding failures.

To address these issues, a blockchain-enhanced random access mechanism named hash access was proposed in [6]. Upon collision, the IoT device is permitted to retransmit only if it satisfies the following condition:

$$\text{Hash}\left(\text{ID} + \text{Timestamp} + \text{Payload}\right) < \text{Target} \tag{1}$$

where ID is the IoT device identifier, Timestamp is the current timestamp, Payload is the data, and Target is the difficulty threshold. Devices failing this condition must defer transmission.

Fig. 1 shows four common circumstances of this blockchain-enhanced random access in untrusted IoT environments. In Fig. 1(a), IoT devices pre-deposit some cryptocurrency on the blockchain, serving as collateral for their following access attempts. In Fig. 1(b), the access is granted, and transactions are recorded in an access contract containing hash proof and service fee. Fig. 1(c) depicts a case of selfish behavior, where a device violating the condition defined by (1) is penalized by an automatic deduction from the device's deposit. Fig. 1(d) shows a scenario where IoT devices perform a random backoff upon they find out their access requests are collided. In these circumstances, blockchain plays a critical role in enhancing coordination, enforcing penalties, and securing access across various subnets.

III. Rogue's Dilemma

In untrusted IoT environments, selfish or rogue devices may disrupt the random access protocol by violating its rules, and thus create the so-called rogue's dilemma.

Definition 1 (Rogue's Dilemma)**.** In wireless access networks, IoT devices are expected to comply with a random backoff rule when attempting to access the channel. However, when IoT devices originate from different operators and thus lack mutual trust, selfish devices may violate the protocol by intentionally modifying their backoff values. Such behavior jeopardizes the performance and fairness experienced by honest devices, and potentially triggers a cascading effect in which all devices adopt selfish access strategies. Ultimately, this leads to significant degradation in overall IoT network performance.

To analyze the impact of this dilemma on individual IoT devices, we begin with a simple setting employing the following assumptions. First, only a single uplink transmission channel is available, meaning that all IoT nodes share the same channel. Second, a collision occurs and data transmission fails if multiple devices transmit simultaneously over the same

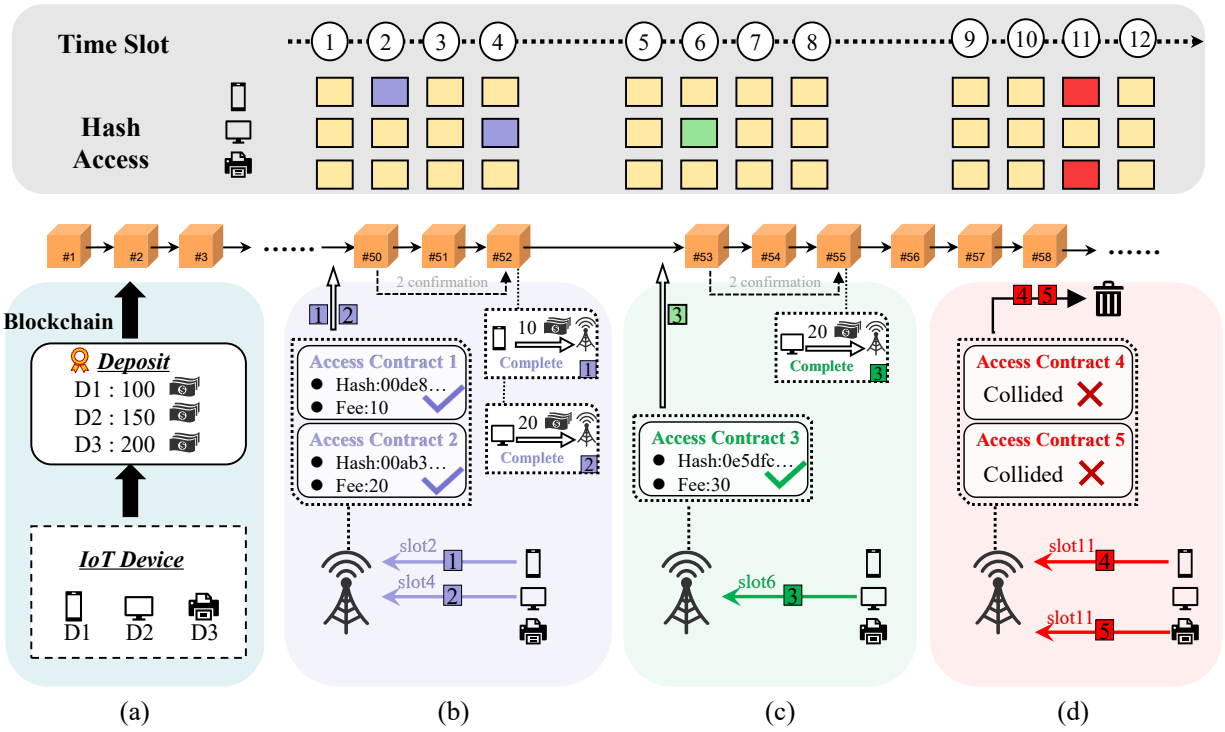


Fig. 1. Blockchain-enhanced random access.

channel. Third, the analysis is restricted to the case in which only two device nodes attempt to transmit within a single time slot.

Accordingly, consider two IoT devices, D_1 and D_2, in the wireless access network. For each IoT device at a given time slot, the set of available strategies is $S_i : \{T; W\}$, $i = 1, 2$, where T denotes transmitting data and W represents waiting or not transmitting. A successful data transmission occurs if exactly one device selects strategy T in a given time slot. If both devices select the strategy T, a collision occurs and the transmission fails. If both choose strategy W, the channel remains idle. Further, assume that a successful transmission yields a reward, denoted by R, $R > 0$. Attempting to transmit incurs a transmission cost, denoted by c_p, $c_p > 0$. If the data is not successfully transmitted in the current time slot, either due to a failed transmission or a decision to wait, a time cost is still incurred, denoted by c_t, $c_t > 0$. Based on the above discussions, we can characterize the behaviors and corresponding payoffs of both honest and rogue IoT devices operating in untrusted environments. The resulting payoff matrix is presented in Table I.

IV. Performance Analysis of Random Access with Rogue Nodes

A. Analysis under Static Game Scenario

Let us first consider a static game scenario where both IoT devices are assumed to be fully rational. Based on the definitions in Section III, we can characterize the game between two IoT nodes as follows.

- In the case of $R - c_p > -c_t$, both IoT devices satisfy $R - c_p > -c_p - c_t$ at the same time, and the rational device will choose the option that yields the higher payoff, i.e., $R - c_p$. This static game obviously has two Nash equilibrium points, i.e., (T, W) and (W, T).
- In the case of $R - c_p \leq -c_t$, the maximum achievable payoff for both IoT devices is $-c_t$ and the final Nash equilibrium point is (W, W). However, since neither IoT device chooses to transmit data in this situation, the channel remains unused, resulting in a significant waste of network resources and greater overall loss. Therefore, this outcome is considered impractical in real-world scenarios.

Therefore, only the case of $R - c_p > -c_t$ is meaningful in the context of fully rational decision-making. Nevertheless, IoT devices do not always behave rationally in practical settings.

	T	W
T	$-c_p - c_t,\ -c_p - c_t$	$R - c_p,\ -c_t$
W	$-c_t,\ R - c_p$	$-c_t,\ -c_t$

Table I: Payoff matrix for two IoT devices under given assumptions.

Factors such as hardware design flaws, varying operator strategies, and selfish access behaviors can prevent devices from consistently choosing optimal strategies. To better show the performance under real-world dynamics, the following subsections explore these issues through mixed-strategy analysis and evolutionary game theory.

B. Analysis under Mixed-Strategy Scenario

We consider a scenario where IoT devices fully trust each other, which is typically applicable within subnets managed by the same operator or produced by the same device manufacturer. In this case, all IoT devices strictly follow the access policy and adopt the transmission probability $p_1 = p_2 = p$ set by the system. At this point, both devices have identical utility functions, denoted by $u_1 = u_2 = u$. Let $u(T)$ denote the expected utility of a device choosing strategy T and $u(W)$ represent the utility when choosing strategy W, which can be expressed as

$$\begin{aligned} u(T) &= (-c_p - c_t)p + (R - c_p)(1 - p) \\ &= R - c_p - (R + c_t)\,p, \end{aligned} \tag{2}$$

$$u(W) = -c_t p + (1 - p)(-c_t) = -c_t. \tag{3}$$

According to (2) and (3), the average expected utility of an IoT device under the mixed-strategy u_{trust} is

$$\begin{aligned} u_{\text{trust}} &= pu(T) + (1 - p)u(W) \\ &= -(R + c_t)p^2 + (R - c_p + c_t)p - c_t. \end{aligned} \tag{4}$$

Given that $R,\ c_p,\ c_t > 0$ and $R - c_p + c_t > 0$, there must exist $p = p^*$ such that the average expected utility u_{trust} is maximized and p^* can be derived as

$$p^* = \frac{R - c_p + c_t}{2(R + c_t)}. \tag{5}$$

In this case, the average expected utility u_{trust} is

$$u_{\text{trust}} \overset{p=p^*}{=} \frac{(R - c_p + c_t)^2}{4(R + c_t)} - c_t. \tag{6}$$

Next, we consider a scenario involving selfish IoT device access. In this scenario, a selfish device chooses the most advantageous strategy for itself after observing the decisions made by honest devices. This situation often arises when IoT devices belong to different operators or manufacturers and thus lack mutual trust. Additionally, certain devices may behave selfishly or maliciously to gain extra benefits. This interaction can be modeled as a two-stage game. Assume the honest device transmits with probability $p_{\text{honest}} = p^*$, and the selfish device deviates from the access policy and privately selects a transmission probability p_{selfish}, which can maximize its own benefit. The selfish device's expected payoffs when the honest device chooses T or W can be expressed as

$$u_{\text{selfish}}(T) = \frac{R - c_p - c_t}{2},\ u_{\text{selfish}}(W) = -c_t, \tag{7}$$

repectively. The average utility of the selfish device can be expressed as

$$u_{\text{selfish}} = \frac{R - c_p + c_t}{2} p_{\text{selfish}} - c_t. \tag{8}$$

Since $R - c_p + c_t > 0$, the selfish device maximizes its benefit by letting $p_{\text{selfish}} = 1$, resulting in

$$u_{\text{selfish}} \overset{p_{\text{selfish}}=1}{=} \frac{R - c_p - c_t}{2}. \tag{9}$$

Furthermore, the selfish behaviors directly impact the payoffs of honest IoT devices. When a selfish device privately increases its transmission probability, the utilities of honest devices are consequently adjusted as

$$u_{\text{honest}}(T) = -c_p - c_t,\ u_{\text{honest}}(W) = -c_t, \tag{10}$$

and

$$u_{\text{honest}} = -\frac{R - c_p + c_t}{2(R + c_t)} c_p - c_t. \tag{11}$$

Comparing the three cases, we find that $u_{\text{honest}} < u_{\text{trust}} < u_{\text{selfish}}$. This shows that unregulated selfish behavior leads to higher individual payoffs at the cost of honest devices. As a result, devices gradually adopt selfish strategy, which causes frequent collisions and eventual system breakdown. To address this, blockchain-enhanced access introduces mechanisms to identify and penalize selfish nodes using pre-deposited stakes and smart contracts. When penalties are set at an appropriate level, rational nodes are motivated to follow the access rules and thus can prevent the system from collapsing into the rogue's dilemma.

C. Long-run Performance Analysis

While the previous mixed-strategy game analysis reveals the equilibrium behavior of honest and selfish devices at a fixed point, real-world IoT networks often involve dynamic interactions and adaptation over time. Device strategies may evolve in response to observed outcomes, network conditions, or changes in the environment. To better capture these dynamic processes and understand how access strategies develop in the

long run, we now adopt an evolutionary game perspective. Consider that IoT devices rely on pre-installed firmware and operate with limited sensing capabilities, and replicator dynamic is employed to model the strategy adaptation process across heterogeneous IoT devices. Note that while IoT devices may observe others' behavior via spectrum sensing modules [15], the limited coverage prevents them from monitoring the full network status.

First, we analyze the strategy evolution without incorporating penalty mechanisms. Let $\boldsymbol{p}_j$ denote the vector of initial transmission probabilities chosen by all N IoT devices at time slot j, $j \in \mathbb{Z}_0^+$, which is expressed as

$$\boldsymbol{p}_j = \left(p_{j0}, p_{j1}, \cdots, p_{ji}, \cdots, p_{j(N-1)}\right), \tag{12}$$

where $i = 0, 1, ..., N-1$ denotes the index of IoT devices and $p_{ji} \in [0, 1]$. It is assumed that each IoT device updates its strategy once per time slot. Given this, the expected benefits of node i adopting the transmission or waiting strategy at time slot j are given by

$$\begin{aligned} u_T(p_{ji}) &= (-c_p - c_t)(1 - \Theta_{ji}) + (R - c_p)\Theta_{ji} \\ &= -c_p - c_t + (R + c_t)\Theta_{ji}, \end{aligned} \tag{13}$$

and

$$u_W(p_{ji}) = -c_t p_{ji} + (1 - p_{ji})(-c_t) = -c_t, \tag{14}$$

respectively, where $\Theta_{ji} = \prod_{k \neq i} \left(1 - p_{(j-1)k}\right)$. According to (13) and (14), the average expected utility u_{avg} for IoT device i at time slot j can be derived as

$$\begin{aligned} u_{\text{avg}}(p_{ji}) &= p_{ji} u_T(p_{ji}) + (1 - p_{ji}) u_W(p_{ji}) \\ &= -c_p p_{ji} - c_t + p_{ji}(R + c_t)\Theta_{ji}. \end{aligned} \tag{15}$$

It is important to note that Θ_{ji} in (13) and (14) is not constant over time. In each round, IoT devices update their strategies based on the strategies adopted by others in the previous round. To simplify the analysis, we assume that all IoT devices simultaneously adjust their strategies based on the observed strategy from the previous time slot. We assume that all IoT devices have bounded rationality and they initially adopt the same transmission probability defined by the system, i.e., p_{ji} in (12) is identical across all i. When one device, e.g., the node i, updates its strategy, we assume all other devices keep their strategies unchanged during that update. Thus, for simplicity, from the perspective of the node i, we let $\tilde{p}_{j-1} = p_{(j-1)k}$ for all $k \neq i$. Based on this assumption, the replicator dynamic equation for this evolutionary game from the perspective of the node i can be expressed as

$$F(\tilde{p}_j) = \tilde{p}_j - \tilde{p}_{j-1} = \tilde{p}_j \left(u_T(\tilde{p}_j) - u_{\text{avg}}(\tilde{p}_j)\right). \tag{16}$$

Under these assumptions, (16) can be further simplified as

$$F(\tilde{p}_j) = \tilde{p}_j \left(\tilde{p}_j - 1\right)\left(c_p - (R + c_t)(1 - \tilde{p}_{j-1})^{N-1}\right). \tag{17}$$

Letting $F(\tilde{p}_j) = \tilde{p}_j - \tilde{p}_{j-1} = 0$, we obtain the following three equilibrium points of the system:

$$\tilde{p}_j^{*(\text{a})} = 0,\ \tilde{p}_j^{*(\text{b})} = 1,\ \tilde{p}_j^{*(\text{c})} = 1 - \left(\frac{c_p}{R + c_t}\right)^{\frac{1}{N-1}}. \tag{18}$$

The average expected utilities for IoT devices at each equilibrium point can be given by

$$\begin{cases} u_{\text{avg}}(\tilde{p}_j = \tilde{p}_j^{*(\text{a})}) = -c_t, \\ u_{\text{avg}}(\tilde{p}_j = \tilde{p}_j^{*(\text{b})}) = -c_p - c_t, \\ u_{\text{avg}}(\tilde{p}_j = \tilde{p}_j^{*(\text{c})}) = -c_t. \end{cases} \tag{19}$$

However, the solutions obtained by (19) may not be robust, i.e., they may not be the ESS, as the ESS further requires that $F^{'}(\tilde{p}_j) = \frac{\mathrm{d}F(\tilde{p}_j)}{\mathrm{d}\tilde{p}_j^*} < 0$ [16]. Therefore, we further perform a robustness check on these equilibrium points. Using (17), we have

$$\begin{aligned} F^{'}(\tilde{p}_j) = &(2\tilde{p}_j - 1)\left(c_p - (R + c_t)(1 - \tilde{p}_{j-1})^{N-1}\right) \\ &+ \tilde{p}_j \left(\tilde{p}_j - 1\right)(N-1)(R + c_t)(1 - \tilde{p}_{j-1})^{N-2}. \end{aligned} \tag{20}$$

Substituting $\tilde{p}_j$ in (20) with the three equilibrium points as derived by (18), we obtain

$$\begin{cases} F_0^{'} = F^{'}(\tilde{p}_j = \tilde{p}_j^{*(\text{a})}) = R + c_t - c_p > 0, \\ F_1^{'} = F^{'}(\tilde{p}_j = \tilde{p}_j^{*(\text{b})}) = c_p > 0, \\ F_2^{'} = F^{'}(\tilde{p}_j = \tilde{p}_j^{*(\text{c})}) = \Upsilon\left(\tilde{p}_j^{*(\text{c})} - 1\right) < 0, \end{cases} \tag{21}$$

where $\Upsilon = (\frac{c_p}{R + c_t})^{\frac{N-2}{N-1}}(R + c_t)(N-1)\tilde{p}_j^{*(\text{c})}$. According to (21), only the point $\tilde{p}_j = \tilde{p}_j^{*(\text{c})}$ satisfies the condition for an ESS. While the equilibrium points $\tilde{p}_j = \tilde{p}_j^{*(\text{a})}$ and $\tilde{p}_j = \tilde{p}_j^{*(\text{b})}$ are stable under replicator dynamics, any slight deviation from these points may lead the network state to drift away without self-correction and ultimately result in system instability or divergence.

V. Numerical Results

To assess how selfish device behavior affects IoT system performance and validate the results we derive in Section IV, we conduct a number of simulations. Specifically, we use a network consisting of $N = 5$ IoT devices. We set the transmission reward $R = 10$, the transmission cost $c_p = 2$, and the time cost $c_t = 1$. These parameters can be mapped to specific currency units or time measures based on practical deployment scenarios. Additionally, the evolution of the

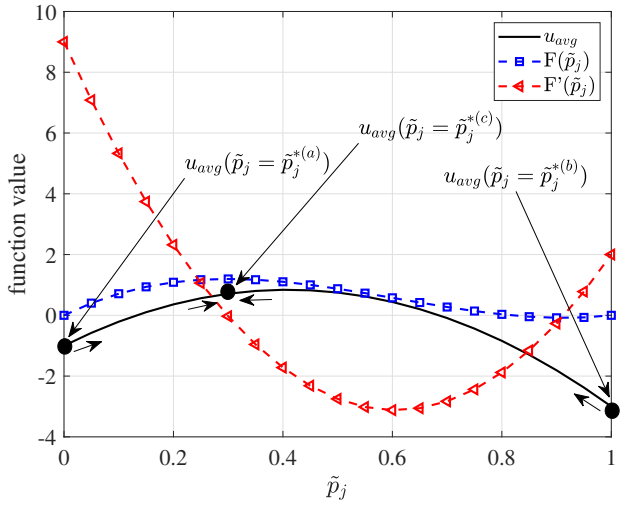


Fig. 2. Trends of $u_{\text{avg}}(\tilde{p}_j)$, $F(\tilde{p}_j)$ and $F^{'}(\tilde{p}_j)$ under varying $\tilde{p}_j$.

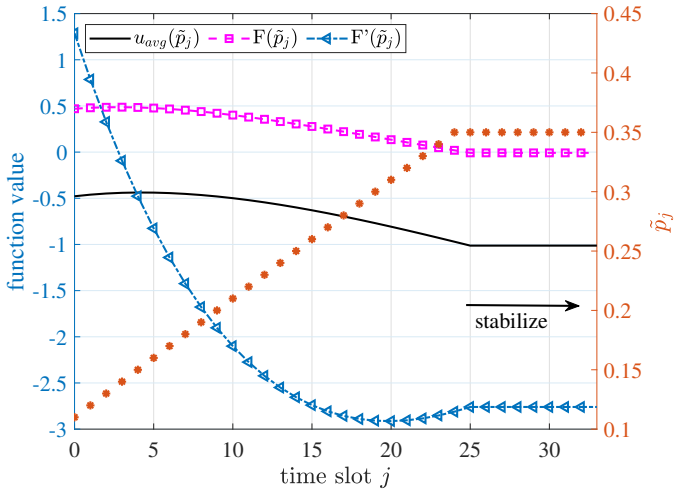


Fig. 3. $u_{\text{avg}}(\tilde{p}_j)$, $F(\tilde{p}_j)$ and $F^{'}(\tilde{p}_j)$ under small initial transmission probability.

transmission probability $\tilde{p}_j$ follows a discrete-time replicator dynamic with a step size of 0.005.

Fig. 2 illustrates the three equilibrium points derived in (18). As derived by our analysis, the equilibrium point $\tilde{p}_j = \tilde{p}_j^{*(c)}$ achieves the highest utility and also serves as an ESS. The other two points, $\tilde{p}_j = \tilde{p}_j^{*(a)}$ and $\tilde{p}_j = \tilde{p}_j^{*(b)}$, are unstable and tend to drift toward the ESS at $\tilde{p}_j = \tilde{p}_j^{*(c)}$. This behavior confirms the validity of our findings.

Next, Fig. 3 shows the evolution of transmission probability, average expected payoff, replicator dynamics function, and its derivative for devices in an untrusted wireless environment, starting from an initial condition of $\boldsymbol{p}_j = (0.1, 0.1, 0.1, 0.1, 0.1)$. As the figure illustrates, the transmission probability $\tilde{p}_j$ increases rapidly and settles at a stable value $\tilde{p}^*$. This reflects how IoT devices tend to raise their transmission probabilities when initially low, aiming to improve individual payoffs. The average expected benefit $u_{\text{avg}}(\tilde{p}_j)$ drops slightly at first as $\tilde{p}_j$ increases, then levels off. This indicates that while higher transmission probabilities can boost the chance of successful access, the overall benefit is limited by a higher risk of collisions. The replicator dynamic function $F(\tilde{p}_j)$ decreases over time and converges to zero. This indicates that $\tilde{p}_j$ is still below the ESS. As a result, the nodes' transmission probability continues to rise. The derivative $F'(\tilde{p}_j)$ confirms that the equilibrium point meets the ESS condition.

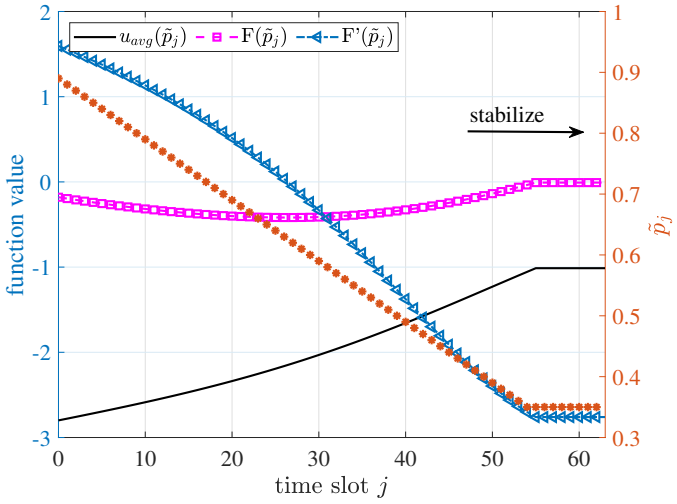


Fig. 4. $u_{\text{avg}}(\tilde{p}_j)$, $F(\tilde{p}_j)$ and $F^{'}(\tilde{p}_j)$ under high initial transmission probability.

Furthermore, Fig. 4 shows the case where the initial transmission probability is high ($\boldsymbol{p}_j = (0.9, 0.9, 0.9, 0.9, 0.9)$). Here, $\tilde{p}_j$ decreases over time and stabilizes at the same $\tilde{p}^*$ as in the previous case. This shows that when the initial access probability is too high, frequent collisions drive nodes to lower their transmission rates. The average expected utility $u_{\text{avg}}(\tilde{p}_j)$ rises as $\tilde{p}_j$ decreases and eventually converges to $-c_t$. This reflects improved efficiency as contention subsides. The replicator dynamic function $F(\tilde{p}_j)$ starts negative and rises to zero, corresponding to $\tilde{p}_j > \tilde{p}_j^*$, and $F'(\tilde{p}_j)$ becomes negative, again confirming convergence to the ESS.

Comparing Fig. 3 and Fig. 4, we can see that regardless of initial conditions, the system evolves toward the same unique ESS, $\tilde{p}_j^{*(c)} = 1 - \left(\frac{c_p}{R+c_t}\right)^{\frac{1}{N-1}} \approx 0.347$. At this point, the average expected utility for each node converges to $-c_t$. This outcome is driven by each node seeking to maximize its

own payoff, i.e., if $\tilde{p}_j$ is too low, increasing it yields better returns; if $\tilde{p}_j$ is too high, collisions reduce the benefit and thus encourage nodes to back off. The replicator dynamics mechanism steers strategies to this stable, locally optimal point. However, as indicated in (19), the average expected payoff remains negative, regardless of the equilibrium. This demonstrates that, without effective penalties for selfish behavior, such as the smart-contract-based penalties enforced by the blockchain, this evolutionary can finally lead to significant network performance degradation.

These results highlight the need for IoT manufacturers and operators to design robust, well-calibrated protocols that prioritize overall network benefit. Our findings show that allowing selfish behavior or blind imitation leads to negative returns, and may further push the system as a whole into the rogue's dilemma. By contrast, blockchain-enhanced random access protocols address this challenge by restricting manipulation of transmission probabilities and help to protect the interests of all devices in the network.

VI. Conclusion

In this paper, we explored blockchain-enhanced random access in untrusted IoT environments from a game-theoretic perspective. We first analyzed the rogue's dilemma, where selfish devices deviate from predefined access rules to gain individual benefits at the cost of overall fairness and efficiency. To characterize such strategic interactions, we developed a two-device mixed-strategy game model and further extended it to a multi-device scenario using evolutionary game theory and replicator dynamics. Using these models, we uncovered the conditions for achieving ESS under bounded rationality. Numerical results showed that unregulated selfish behavior leads to severe performance degradation and unstable access dynamics, whereas blockchain-based penalties are proven to effectively deter such behavior and promote system stability. The analysis and findings in this paper can provide theoretical insights and practical guidance for designing robust and trusted access protocols in future communication networks.

References

[1] M. Nassereddine and A. Khang, "Applications of Internet of Things (IoT) in smart cities," in *Advanced IoT Technologies and Applications in the Industry 4.0 Digital Economy*. CRC Press, Feb. 2024, pp. 109–136.

[2] E. U. Haque, A. Shah, J. Iqbal, S. S. Ullah, R. Alroobaea, and S. Hussain, "A scalable blockchain based framework for efficient IoT data management using lightweight consensus," *Sci. Rep.*, vol. 14, no. 1, p. 7841, Apr. 2024.

[3] S. F. Ahmed, M. S. B. Alam, S. Afrin, S. J. Rafa, S. B. Taher, M. Kabir, S. Muyeen, and A. H. Gandomi, "Toward a secure 5G-enabled Internet of Things: A survey on requirements, privacy, security, challenges, and opportunities," *IEEE Access*, vol. 12, pp. 13 125–13 145, Jan. 2024.

[4] X. Chen, W. Wu, L. Li, and F. Ji, "LLM-empowered IoT for 6G networks: Architecture, challenges, and solutions," *IEEE Internet Things Mag.*, pp. 1–8, Feb. 2025, early Access.

[5] C. Fernandez-Gago, D. Ferraris, R. Roman, and J. Lopez, "Trust interoperability in the Internet of Things," *Internet Things*, vol. 26, p. 101226, Jul. 2024.

[6] X. Ling, Y. Le, J. Wang, and Z. Ding, "Hash Access: Trustworthy grant-free IoT access enabled by blockchain radio access network," *IEEE Network*, vol. 34, no. 1, pp. 54–61, Jan. 2020.

[7] X. Ling, J. Wang, T. Bouchoucha, B. C. Levy, and Z. Ding, "Blockchain radio access network (B-RAN): Towards decentralized secure radio access paradigm," *IEEE Access*, vol. 7, pp. 9714–9723, Jan. 2019.

[8] S. Nakamoto, "Bitcoin: A peer-to-peer electronic cash system," Tech. Rep., Oct. 2008. [Online]. Available: https://bitcoin.org/bitcoin.pdf

[9] Q. Chen, D. Li, and L. Wang, "Blockchain technology for enhancing network security," *J. Ind. Eng. Appl. Sci.*, vol. 2, no. 4, pp. 22–28, Aug. 2024.

[10] I. Abellán Álvarez, V. Gramlich, and J. Sedlmeir, "Unsealing the secrets of blockchain consensus: A systematic comparison of the formal security of proof-of-work and proof-of-stake," in *Proc. 39th ACM/SIGAPP Symp. Appl. Comput.*, New York, NY, US, May 2024, pp. 278–287.

[11] X. Ling, Y. Le, J. Wang, Y. Huang, and X. You, "Trust and trustworthiness in information and communications technologies," *IEEE Wireless Commun.*, vol. 32, no. 2, pp. 84–92, Apr. 2025.

[12] A. B. MacKenzie and S. B. Wicker, "Selfish users in ALOHA: a game-theoretic approach," in *Proc. 54th IEEE Veh. Technol. Conf. (VTC' 01)*, Atlantic City, NJ, US, Oct. 2001, pp. 1354–1357.

[13] K. Kotobi and S. G. Bilén, "Blockchain-enabled spectrum access in cognitive radio networks," in *Proc. 2017 Wireless Telecommun. Symp. (WTS' 17)*, Chicago, IL, US, Apr. 2017, pp. 1–6.

[14] N. Abramson, "The ALOHA system: Another alternative for computer communications," in *Proc. 1970 Fall Jt. Comput. Conf. (FJCC' 70)*, Houston, TX, US, Nov. 1970, pp. 281–285.

[15] S. Ziafat, W. Ejaz, and H. Jamal, "Spectrum sensing techniques for cognitive radio networks: Performance analysis," in *Proc. 2011 IEEE MTT-S Int. Microwave Workshop Ser. Intell. Radio Future Pers. Terminals*, Daejeon, KR, Aug. 2011, pp. 1–4.

[16] J. Hofbauer and K. Sigmund, *Evolutionary Games and Population Dynamics*. Cambridge, UK: Cambridge University Press, 1998.